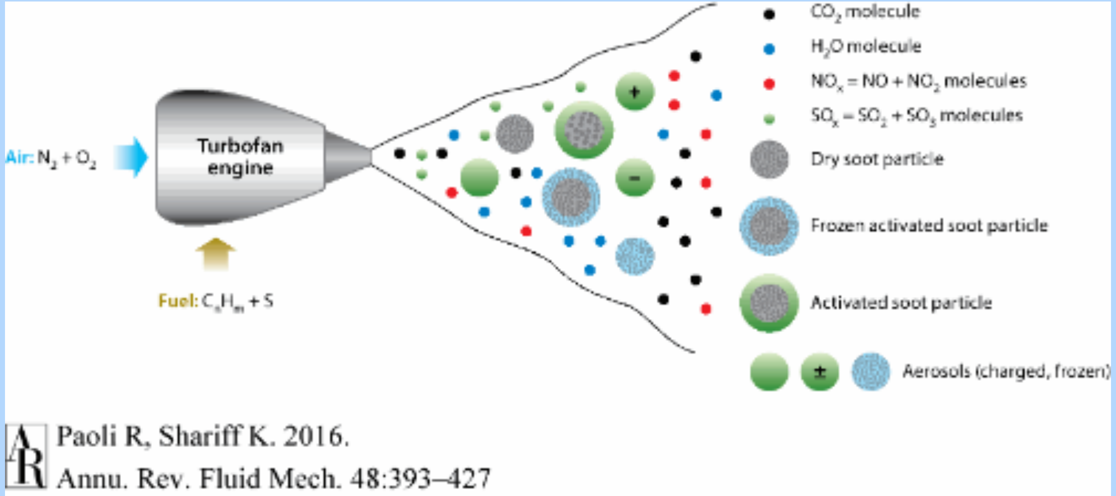
Uncertainty-Aware Contrail Mitigation

Explainable Sequential Planning for Aviation Climate Impact

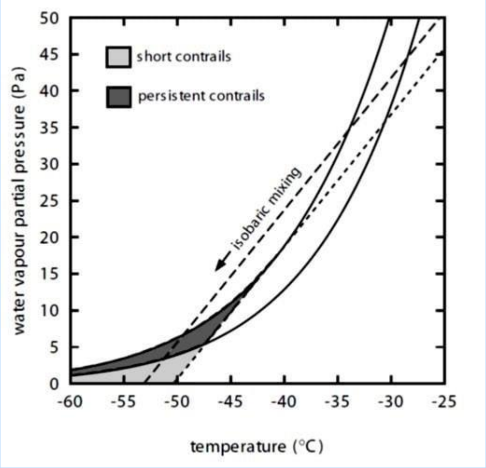
Anika Chawla

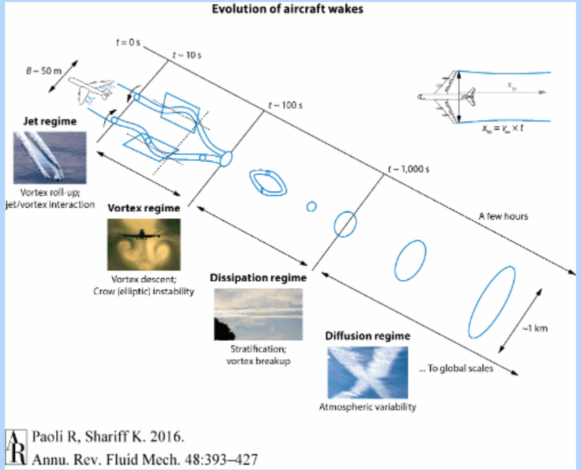
Background - Contrails

- Contrails (condensation trails) are ice clouds formed from aircraft exhaust
- Produced when water vapor condenses and freezes in cold, humid, air - governed by Schmidt-Appleman Criterion
- Aircraft engines emit:
  - CO_2, H_2O, NO_x, SO_x, HC, and soot particles
- Occur upper troposphere and lower stratosphere (9-12 km)
  - Region where long-haul aircraft spend most of their time
- Can persist and spread into contrail cirrus, impacting climate



Paoli R, Shariff K. 2016. Annu. Rev. Fluid Mech. 48:393–427





Paoli R, Shariff K. 2016. Annu. Rev. Fluid Mech. 48:393–427

Contrail Climate Impacts

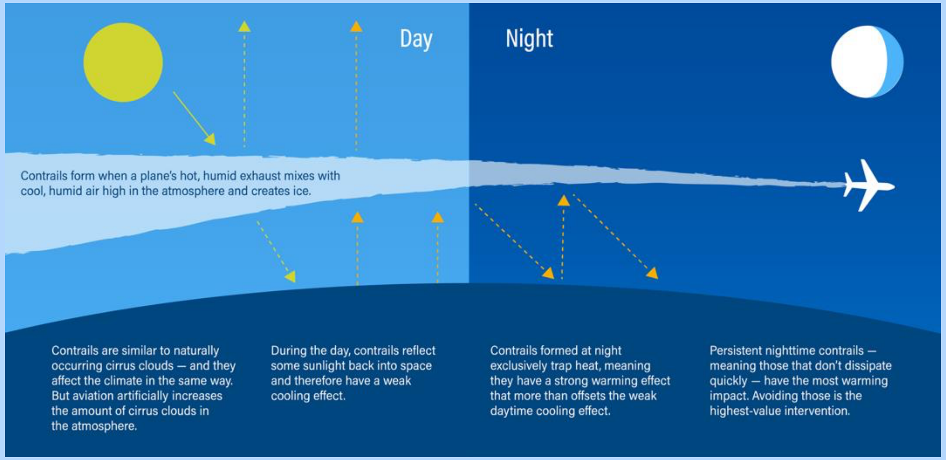
Major contributor to aviation's non-CO2 climate impact

~5% of flights → ~80% of contrail warming

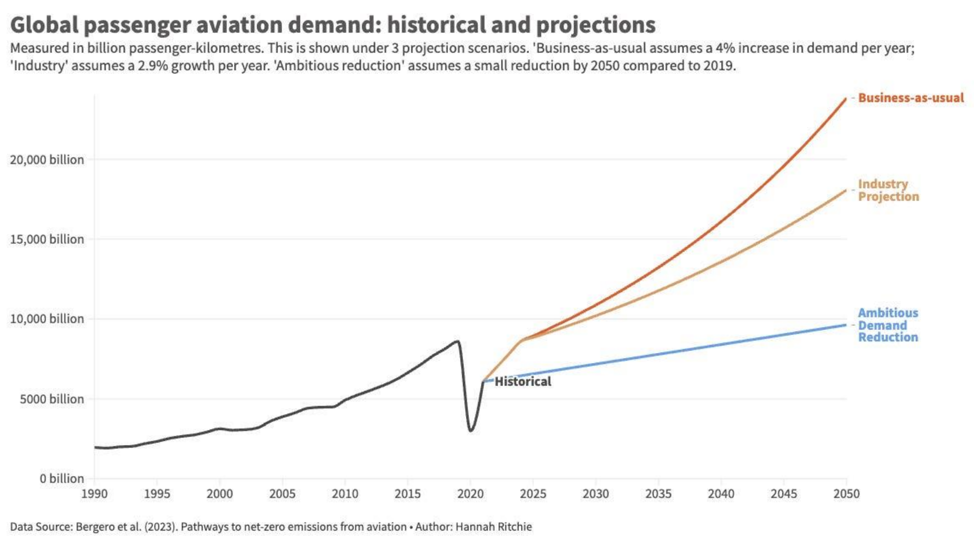
Contrails cause 1-2% of global warming

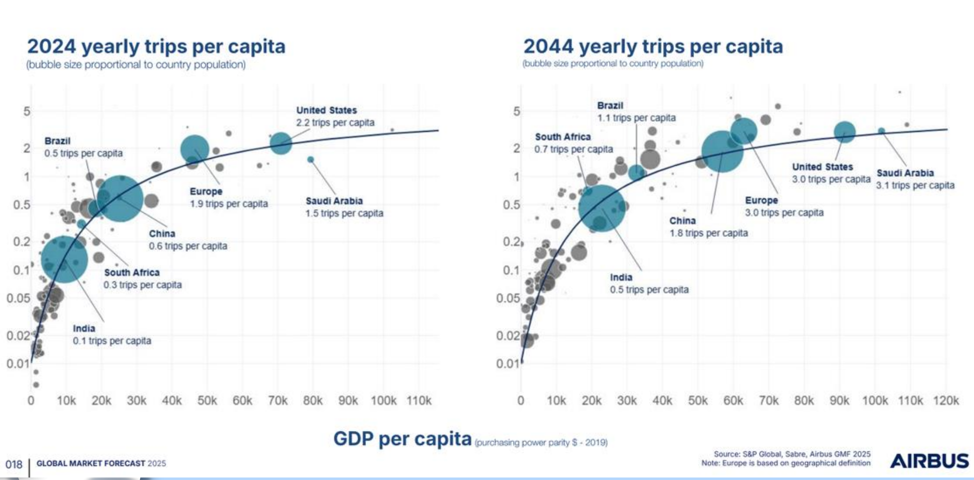
Recent estimates show that contrails cause about the same amount of warming each year as all the CO_2 emitted by all flights from 1940 to 2018

Contrail RF depends on time of day, season, cloud cover, albedo, etc. – in aggregate contrails are strongly warming



Aviation Demand





- Aviation demand is increasing – expected to double by 2050
- Growth especially in long haul routes, which spend more time in contrail-prone region
- Increased flight frequency → more opportunities for contrail formation
- Expansion of long-haul operations → more exposure to ISSR
- Greater airspace density → increased cumulative effects
- Crucial to address aviation's CO_2 and non-CO_2 impacts as industry grows

Contrail Mitigation Approaches

Sustainable Aviation Fuel (SAF)

- Lower aromatic content → reduced soot emissions
- Fewer cloud condensation nuclei → fewer ice crystals
- Blended SAF (up to ~50%) → **~50–70% reduction in soot emissions**
- Leads to smaller, less persistent contrails
- **Challenge:** High cost and limited production scale

New Engine Technology

- More efficient combustion → lower soot and particulate emissions
- Fleet-wide adoption studies show up to **~70% reduction in contrail warming impact**
- Improves baseline emissions across entire operations
- **Challenge:** Slow fleet turnover (decades-long replacement cycles)

Intelligent route planning

- Persistent contrails form Ice super saturated regions, redirecting a small number of flights (5%) can lead to large decrease in contrail climate warming (80%)
- Only $1 average cost of avoiding warming equivalent to one ton of CO_2 – field trails showed a >50% reduction in contrail formation with 1–2% increase in fuel consumption
- *This is a near-term solution where large climate benefits can be reaped with a small economical and logistical cost while using current infrastructure*

Contrail Formation Prediction Challenges

Atmospheric Uncertainty

- ISSRs are thin, intermittent, and poorly resolved
- Strong spatial gradients lead to small errors in humidity lead to large prediction changes
- Upper-tropospheric humidity is among the least accurately measured atmospheric variables
- Radiosonde, reanalysis, and satellite products often disagree significantly

Data Limitations

ERA5 / NWP reanalyses

- Smooth out fine-scale humidity structure
- Systematic biases in relative humidity over ice

Satellite retrievals

- Limited vertical resolution (2D images cannot capture depth well)
- Viewing-angle and cloud-contamination errors
- Difficulty resolving thin ISSR layers

Physical Complexity

- Contrail formation and persistence depend on tightly coupled processes
- Jet + vortex phase dynamics (seconds–minutes)
- Ice nucleation and growth (microphysics)
- Turbulent mixing, shear, and radiative effects (minutes–hours)
- Strong coupling between thermodynamics and fluid dynamics

Modeling Limitations

CoCiP

- Fast, operationally usable
- Highly simplified microphysics and dynamics

APCEMM

- High-fidelity microphysics and evolution
- Too computationally expensive for routing decisions

No existing model has resolution/time-scale for physical accuracy and operational feasibility

Opportunity for Improvement

Existing approaches rely on:

- Deterministic thresholds
- Single-point predictions
- *Do not account for atmospheric uncertainty, model-form error, and data-disagreement*

Result: Suboptimal and potentially overconfident decisions

Key Insight: *Contrail mitigation is fundamentally a decision-making problem under uncertainty and partial observability*

Proposed Solution

Develop an uncertainty-aware explainable framework that integrates:

- Atmospheric data, Physical modeling, Bayesian inference, Sequential decision-making

Goal: Enable robust, risk-aware contrail mitigation strategies

Atmospheric Observations & Forecast Data

NWP ensembles, ERA5, AIRS/IASI humidity, satellite retrievals

→ **CoCiP / pycontrails Simulations** →

Probabilistic Surrogate Model

Bayesian NN / ensemble surrogate

Gaussian-process discrepancy model

Bayesian calibration using atmospheric & satellite data

Outputs:

- Contrail formation probability
- Persistence probability
- Radiative forcing proxy distributions
- Uncertainty decomposition (atmospheric / parametric / model-form)

→ **Surrogate Predictions** →

Sequential Decision Framework

POMDP formulation

Policy proposal for feasible trajectory adjustments

MCTS rollout evaluation

Risk-sensitive objective

Inputs Propagates through planning:

- Atmospheric ensemble variability
- Surrogate uncertainty
- Model discrepancy
- Operational uncertainty

Outputs:

- Recommended trajectory adjustments
- Expected RF reduction
- Decision Confidence

→ **Planner Decisions** →

Evaluation & Validation

Paired-difference experimental design

Baseline vs planner-modified trajectories

Metrics:

- Contrail RF reduction
- Deviation distance/operational cost
- Humidity regime sensitivity
- System-level treatment effects

Validation using atmospheric and satellite datasets

Probabilistic Surrogate Model

CONSTRUCT A **BAYESIAN SURROGATE OF COCIP** THAT CAPTURES:

INPUT UNCERTAINTY (HUMIDITY, TEMPERATURE, WINDS)

PARAMETRIC UNCERTAINTY (MICROPHYSICS, EMISSIONS)

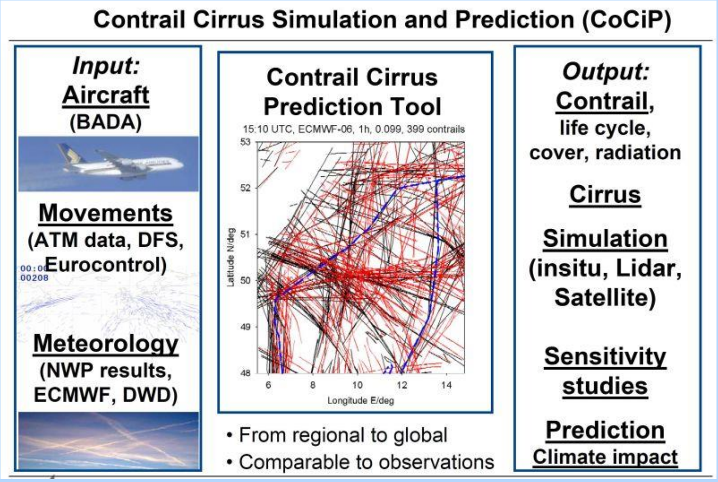
MODEL-FORM DISCREPANCY

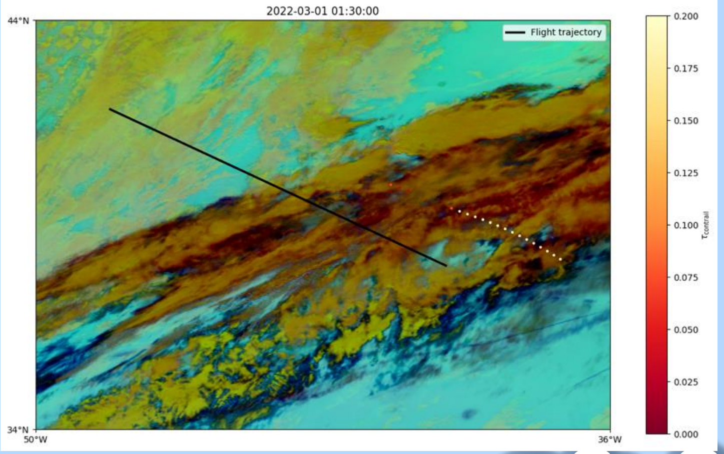
USES:

ENSEMBLE WEATHER DATA

SATELLITE OBSERVATIONS

REANALYSIS DATASETS



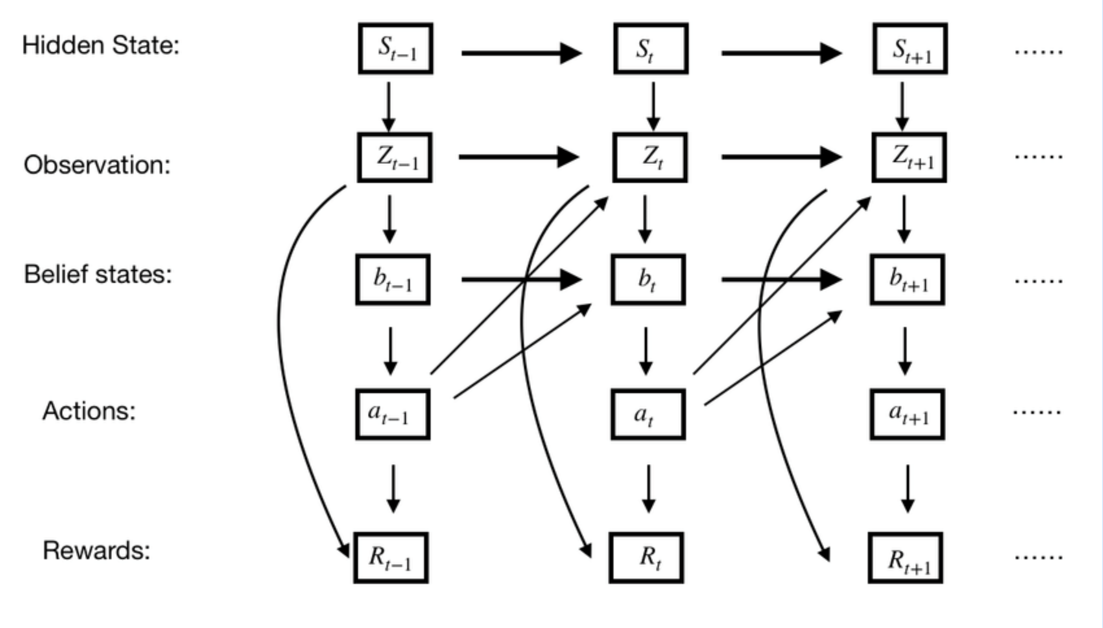


Sequential Planning Framework

Formulate mitigation as a **Partially Observed Markov Decision Process (POMDP)**

Use:

- Monte Carlo Tree Search (MCTS)
- Risk-sensitive objectives (CVaR)
- Optimize **trajectory adjustments over time**, not just isolated points



Explainability & Uncertainty Decomposition

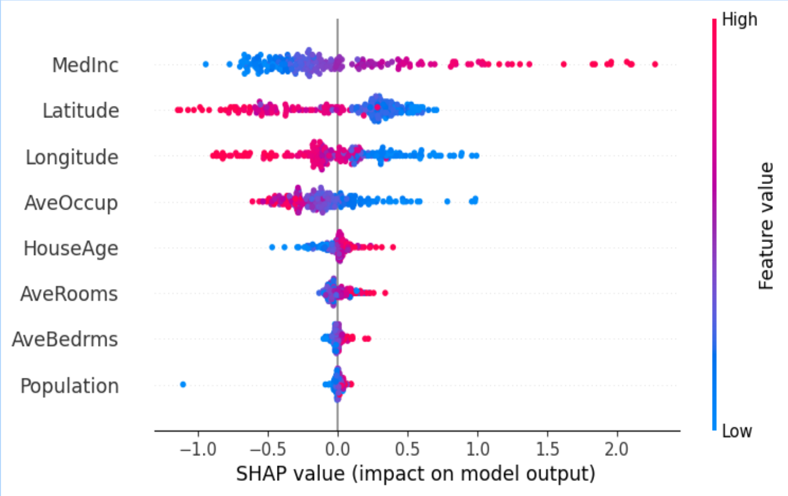
Apply:

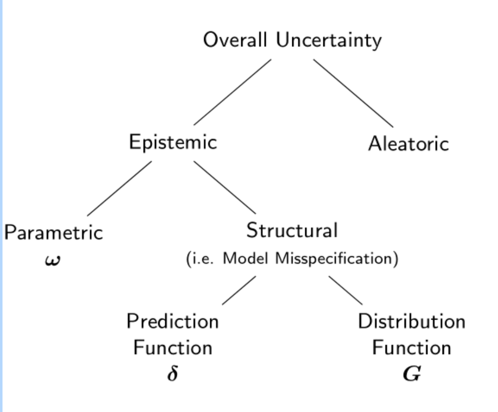
SHAP	Integrated gradients

Decompose uncertainty into:

Atmospheric variability	Model uncertainty	Data limitations

Enables **interpretable decision support**





Evaluation Strategy

Use **paired-difference experimental design**

Compare:

- Baseline vs modified trajectories

Metrics:

- Radiative forcing reduction
- Route deviation cost
- Treatment effects

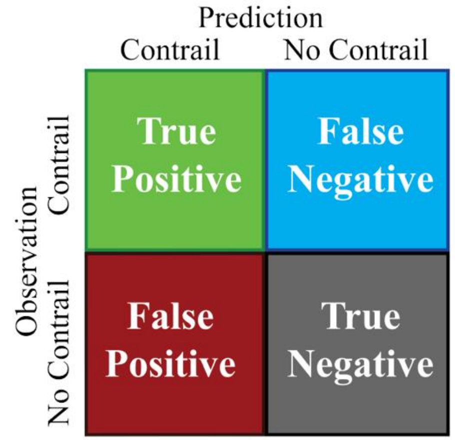


Figure 1.1: Confusion matrix of the outcomes of a binary classifier

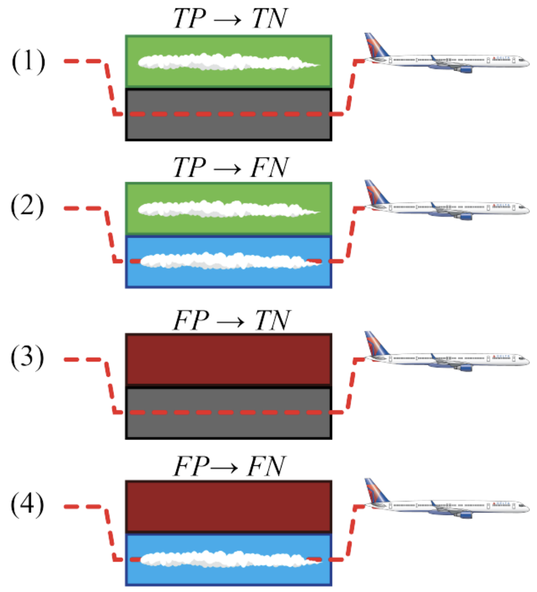


Figure 1.2: Outcomes of contrail avoidance, based on [1]

Expected Benefits/Limitations

Climate: Target high-impact flights for maximum reduction

Operational: Maintain minimal fuel penalties

Decision-making: Improve confidence through uncertainty awareness

Framework applicable to: Climate modeling, Energy systems, environmental decision-making

<u>Challenges and Limitations:</u>

- Persistent uncertainty in humidity measurements
- Structural biases in observational data
- Computational cost of Bayesian methods

Conclusion

Contrails represent a **high-impact, tractable climate problem**

Current approaches are limited by **uncertainty and model gaps**

This work aims to combine:

- ✈ Probabilistic modeling
- ✈ Sequential decision-making
- ✈ Explainable AI